SJAA Ephemeris Bulletin

July 2025

South Saturn

Who doesn't love Saturn? I can't think of any other astronomical object that is immediately recognizable by almost anyone at the eyepiece of a telescope, regardless of their stargazing experience. It's one of my favorite objects to show at outreach events, and it has been creeping into our skies again in the early morning hours.

In this current apparition, Saturn is presenting its southern hemisphere to the Sun. Saturn's axial tilt is about 27 degrees, similar to Earth's tilt of 23 degrees. On both Earth and Saturn, this tilt is what accounts for the seasons. For half of a year one hemisphere is closer to the Sun than the other. Saturn's year is equivalent to 29.5 Earth years, so we're entering a 15-year period of observing Saturn's southern hemisphere.

Before Saturn dominates the night sky again, take some time to follow Joe Fragola's target of the month, featured at right. Better yet, meet us at one of our star parties (calendar on page 2), and see it for yourself! I hope to see you soon under the stars. Keep looking up.

Carl Svensson
Vice-President
Ephemeris Editor

Target of the Month: M16

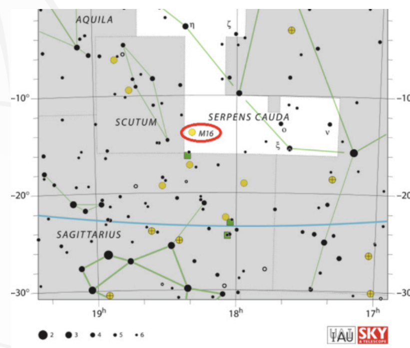
Magnitude	6.4
Size	70 ly by 55 ly
Object Type	Emission Nebula
Constellation	Serpens
Distance	7000 ly
Age	5.5 million years

Messier 16, also known as the “Eagle Nebula,” is a star-forming H-II region containing a young open star cluster. The star cluster was discovered by Swiss astronomer Jean-Philippe de Cheseaux in 1745, but he did not mention the nebula in his notes. Messier 16 was independently discovered by Charles Messier on June 3, 1764.

The Eagle Nebula is also cataloged as IC 4703, while the star cluster has a designation of NGC 6611. The “Pillars of Creation” are towering columns of interstellar gas and dust located within the Eagle Nebula. The largest “pillar” is between 4 and 5 light years high. The Hubble Space Telescope captured a famous image of this region on April 1, 1995.

The open cluster NGC 6611 contains about 460 stars. The brightest members are only 1 to 2 million years old and belong the spectral class O (blue-white supergiants with surface temperatures between 45,000°F and 90,000°F).

You can find the Eagle Nebula in the “tail” portion of the constellation Serpens, the Serpent, near Scutum. It lies about 2.5 degrees west of the bright star Gamma Scuti. Messier 16 is not that impressive through binoculars. The open star cluster will easily be seen, but the nebulosity will not be obvious. Views through a telescope will show the nebulosity and become more rewarding with larger apertures.



Messier 16 location. Image: IAU and Sky & Telescope magazine (Roger Sinnott & Rick Fienberg)

Top: A 38-minute exposure of M16 from 5-22-2025 at RCDO by Joe Fragola. Bottom: M16 location by IAU and Sky and Telescope.

Announcements

There are many volunteer opportunities in SJAA! While we love having seasoned experts in our volunteer ranks, we happily welcome enthusiastic beginners, too. Please do not hesitate to reach out to the SJAA Board (board@sjaa.net) to inquire about opportunities. Below are a few examples of programs that could use help:

School Star Party (SSP) Educating the public about astronomy and related sciences is our mission at SJAA. One of the most effective ways we do this is by hosting star parties for students! Come join the ranks of friendly and knowledgeable volunteers, and share the wonders of the cosmos with elementary through high school aged kids. Send an email to ssp@sjaa.net to get started.

Ephemeris Want to help produce the official publication of the SJAA? A volunteer here would help with finding and creating original content that showcases the club’s activities, and relevant events in the astro community. Play reporter by attending SJAA events, and writing about them in our quarterly newsletter! Send a message to ephemeris@sjaa.net to get started.

A few down-to-Earth needs Do you like working with other SJAA members who share your interest in astronomy? SJAA is looking for volunteer members to take on the role of Volunteer Coordinators (2) and Hands-on Imaging Coordinator. Descriptions of these positions can be found at https://www.sjaa.net/contribute/how-to-volunteer/. Please contact the SJAA Board at board@sjaa.net if you would like additional information about these position or how you can offer your help.

Astro 101 is here for you Did you know that SJAA gives twice-monthly presentations about star gazing, observing equipment, astronomy, and related sciences? It’s true. The Astronomy 101 program hosts these presentations, created and presented by SJAA volunteers. Geared towards beginners, these presentations can teach or remind you about the many branches of amateur astronomy. But, that’s not all! Astro 101 also hosts a periodic, interactive Beginner’s Forum, and even has a comprehensive Telescope Workshop. Learn more and access valuable reference material at https://www.sjaa.net/programs/beginners-astronomy/

San Jose Astronomical Association
SJAA 70 YEARS
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SJAA July 2025

Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
		1 43%	**2** First Quarter Moon 53%	**3** Spica 0.8° N of the Moon Mercury 1.5° S of M44 63%	**4** Venus 2° S of Uranus Henrietta Swan Leavitt's Birthday (1879) 72%	**5** 80%
6 87%	**7** Antares 0.4° N of Moon 93%	**8** 97%	**9** 97%	**10** Full Moon (Buck Moon) 100%	**11** Pluto 0.2° S of Moon 100%	**12** SJAA Guest Speaker 8:00PM SJAA Board Meeting 6:00PM 97%
13 93%	**14** 86%	**15** Imaging SIG 7:30PM Jocelyn Bell Burnell's Birthday (1943) 77%	**16** Saturn 4° S of Moon Neptune 3° S of Moon 67%	**17** Last Quarter Moon 55%	**18** Astro 101 8:15PM In-Town Star Party 9:30PM Yosemite Star Party 44%	**19** Starry Nights 9:30PM Binocular Stargazing 9:15PM Edward Pickering's Birthday (1846) 32%
20 Solar, Loner, Fixit 2:00PM Moon 0.8° N of the Pleiades (M45) 21%	**21** 12%	**22** Jupiter 5° S of Moon 6%	**23** Vera Rubin's Birthday (1928) 1%	**24** New Moon 0%	**25** 2%	**26** Mendoza Ranch Observing 9:30PM Regulus 1.3° S of Moon 5%
27 11%	**28** Peak of Southern Delta Aquariid Meteor Shower Mars 1.3° N of Moon 18%	**29** 27%	**30** Spica 1° N of the Moon 36%	**31** 46%		

Times and dates are subject to change. Please check sjaa.net and www.meetup.com/sj-astronomy for the latest updates.

Dark Sky Window